Mock exam 1

Level: Easy to medium **Marks: 25** **Time: 45 minutes**

- All questions are compulsory.
- The marks for each question are shown against it.
- In Section A choose the one correct option. In the other sections write complete sentences.

Section A: Multiple choice questions

6 x 1 = 6 marks

1. Which of these is a magnetic material? [1]

(a) Nickel **(b)** Wood

(c) Glass **(d)** Plastic

2. Magnets that are found in nature are called [1]

(a) artificial magnets **(b)** lodestones

(c) compasses **(d)** iron filings

3. A bar magnet is placed on a heap of iron filings and lifted. Where do most of the filings stick? [1]

(a) In the middle of the magnet **(b)** Equally all over the magnet

(c) Near the two ends of the magnet **(d)** Only on one end of the magnet

4. A bar magnet hangs freely from a thread. Along which direction does it come to rest? [1]

(a) Any direction **(b)** East-west

(c) It never comes to rest **(d)** North-south

5. The South pole of one magnet is brought near the South pole of another magnet. The two poles will [1]

(a) repel each other **(b)** attract each other

(c) not affect each other **(d)** first attract and then repel

6. **Assertion (A):** A magnet can move a compass needle even when a sheet of cardboard is kept between them. [1]
   Reason (R): The magnetic effect can act through non-magnetic materials.

   (a) Both A and R are true, and R explains A.

   (b) Both A and R are true, but R does not explain A.

   (c) A is true, but R is false.

   (d) A is false, but R is true.

Section B: Very short answer questions

2 x 2 = 4 marks

7. Name the two poles of a magnet. Can a magnet have only one pole? [2]

8. Write any two things that you should not do to a magnet. [2]

Section C: Short answer questions

2 x 3 = 6 marks

9. Meena brings a magnet near an iron nail, a plastic scale and a glass tumbler. [3]
   (a) Which object sticks to the magnet?
   (b) Which objects are made of non-magnetic materials?
   (c) What is a magnetic material?

10. You are standing in a playground with a magnetic compass. Write three steps to find the directions with it. [3]

Section D: Long answer question

1 x 5 = 5 marks

11. Arun places a bar magnet on a few round pencils. He brings a second bar magnet near it, without touching. [5]
   (a) What happens when he brings a North pole near the North pole of the first magnet?
   (b) What happens when he brings a South pole near the North pole of the first magnet?
   (c) What are these two effects called?
   (d) He now replaces the magnet on the pencils with an iron bar. What will he see when he brings each pole of his magnet near it? What does this tell him?

Section E: Case-based question

1 x 4 = 4 marks

12. Read the passage and answer the questions. [4]

Reshma sets up three games at her school fair. In Game 1, steel balls lie inside a maze on a cardboard tray, and players move a magnet below the tray. In Game 2, a steel paper clip lies in a glass of water. In Game 3, two matchbox cars each carry a bar magnet, with the South poles facing each other.

(i) In Game 1, why do the steel balls move although the magnet does not touch them?

(ii) In Game 2, how can a player take out the clip without wetting the fingers or the magnet?

(iii) In Game 3, what happens when the two cars are brought close and let go? Why?

(iv) What one change will make the two cars rush towards each other?